

Efficient Single Image Super-Resolution

Using Frequency-Inspired Optimization in Transformer

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Introduction to Super-Resolution

- **Super-Resolution (SR)** aims to reconstruct a high-resolution (HR) image from a low-resolution (LR) input.
- Transformer-based approaches have shown promising results but struggle with high-frequency details.
- This paper introduces the **CRAFT model** to combine CNN and Transformer benefits.

Abstract

- Transformers have been useful to extract long range dependencies.
- However, it is found that transformer structures are better at capturing low-frequency information, and have limited capacity in constructing high-frequency representations when compared to their convolutional counterparts
- The paper proposes a **CRAFT** transformer which is claimed to carry advantages of transformer and convolutional structures.
- Has three components namely **HFERB** for high-freq enhancement, **SRWAB** for global information and **HFB** for refining representation.
- Introduction of frequency-guided post-training quantization (PTQ) method aimed at enhancing CRAFT's efficiency, which also incorporate adaptive dual clipping and boundary refinement.
- PTQ strategy also extended to function as a general quantization method for transformer-based SISR techniques

- Introduces the cross-refinement adaptive feature modulation transformer (CRAFT), an innovative model that effectively harnesses the strengths of both CNNs and transformers.
- Use of a building block known as CRFB - Cross-Refinement Fusion Blocks.
- Multiple CRFBs form a RCRFG - Residual Cross-Refinement Fusion Group.
- These CRFBs integrate high-frequency information with long-range dependencies.

Residual Cross-Refinement Fusion Group (RCRFG)

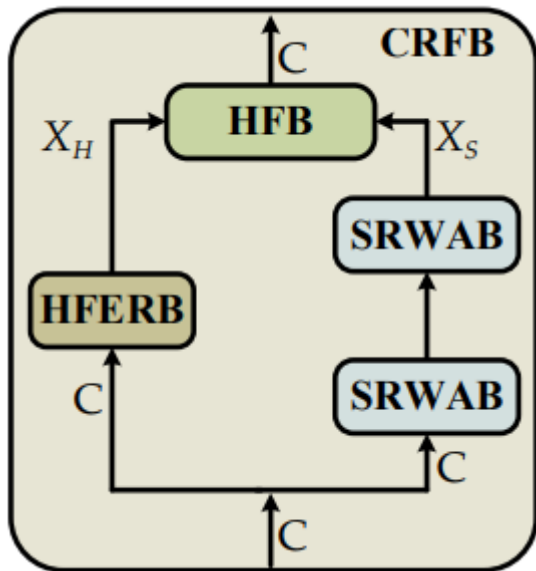
- RCRFG is the core module of CRAFT, responsible for feature enhancement.
- It consists of multiple Cross-Refinement Fusion Blocks (CRFBs).

Block

Consists of

- High-Frequency Enhancement Residual Block (HFERB)
- Shift Rectangle Window Attention Block (SRWAB)
- Hybrid Fusion Block (HFB)

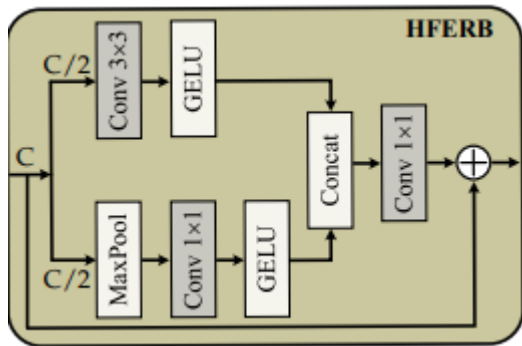
CRFB



HFERB: High-Frequency Enhancement Residual Block

- To handle and construct high frequency data, an image feature representation is used for the HFERB.
- It consists of multiple Convolutional and Max-Pooling blocks.
- It comprises the local feature extraction (LFE) branch and the high-frequency enhancement (HFE) branch.
- LFE using 3x3 Convolution block and GELU.
- HFE using max pool, 1x1 convolution and GELU.
- Concatination and use of Skip Connection for stability and gradient.

HFERB



SRWAB: Shift Rectangle Window Attention Block

- To handle long range dependencies.
- Input divided into rectangles of $H \times W \frac{1}{rh \times rw}$
- It comprises query, key and value

$$Q_i = X_i W, \quad K_i = X_i W_K, \quad V_i = X_i W_V$$

for each i th rectangle

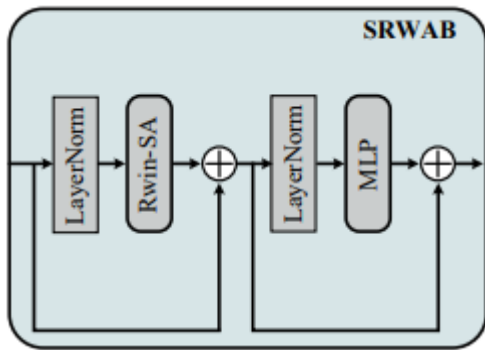
- Self Attention as:

$$\text{Attention}(Q_i, K_i, V_i) = \text{Softmax} \left(\frac{Q_i K_i^T}{\sqrt{d}} + B \right) V_i$$

where B is the dynamic relative position encoding.

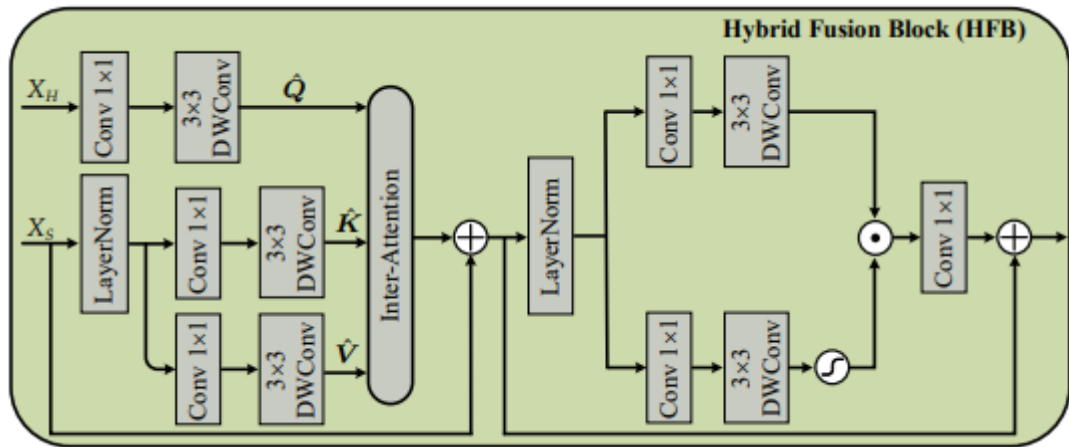
- To capture information from different axes, use two types of rectangle windows: horizontal and vertical windows.
- Split the attention heads into two equal groups and compute the self-attention within each group separately

SRWAB



HFB: Hybrid Fusion Block

- To better integrate the merits of CNN and transformer (HFERB and SRWAB), a hybrid fusion block (HFB) is used.
- Formulate the output of HFERB as the high-frequency prior query and the output of SRWAB as key, value and calculate the inter-attention to refine the global features which are obtained from SRWAB.
- To overcome loss of information through the channel an inter-attention based on the channel dimension to explore channel dependencies is used.
- A 1×1 convolutional layer followed by a 3×3 depth-wise convolutional layer to generate the high-frequency query $Q \in \mathbb{R}^{H \times W \times C}$ based on the output of HFERB
- As to the output of SRWAB normalize the features by the LayerNorm layer and then use the same operation as the query Q to get the key $K \in \mathbb{R}^{H \times W \times C}$ and the value $V \in \mathbb{R}^{H \times W \times C}$.
- Inter-Attention as $\text{Attention}(\hat{Q}, \hat{K}, \hat{V}) = \text{Softmax}\left(\frac{\hat{Q}\hat{K}^T}{\alpha}\right) \hat{V}$



Frequency-Guided PTQ Strategy

- Transformer-based SISR models are computationally expensive.
- Quantization reduces storage and inference cost by lowering precision (e.g., 8-bit, 4-bit) for weights and activations.
- Standard Quantization-Aware Training (QAT) requires extensive retraining, which is infeasible for many practical scenarios.
- PTQ allows post-training quantization without retraining, making it more efficient.
- * PTQ directly to transformers causes severe performance degradation due to high dynamic range in activations (difficult to quantize).
- * Loss of high-frequency details, essential for super-resolution.

Post-Training Quantization (PTQ)

Purpose: Dynamically optimizes clipping range for activations.

- Computes min/max values for activations.
- Uses histogram-based refinement to determine optimal clipping bounds.
- Optimizes clipping range iteratively to minimize quantization loss.
- b is the no. of bits to be used for quantization.

Mathematical Formulation:

$$\text{scale} = \frac{u - l}{2^b - 1}, \quad \text{zero point} = \text{clip} \left(\left\lfloor \frac{-l}{\text{scale}} \right\rfloor, 0, 2^b - 1 \right) \quad (1)$$

Quantization Process:

$$X_i^{\text{int}} = \text{clip} \left(\left\lfloor \frac{X_i}{\text{scale}} \right\rfloor + \text{zero point}, 0, 2^b - 1 \right) \quad (2)$$

- Converts floating-point input X_i into an integer representation X_i^{int} .
- Uses a scaling factor and zero point for precise mapping.

Adaptive Dual Clipping (ADC)

Purpose: Adaptive Dual Clipping (ADC) is a quantization technique designed to optimize the clipping bounds $\{l, u\}$ for minimizing quantization error. By adaptively adjusting these bounds, it improves the accuracy of low-bit quantization.

Methodology:

- Initialize clipping bounds l and u from the minimum and maximum values of the quantization layer F .
- Compute frequency-guided criteria using the *Frequency-Guided Criteria Measuring Processing (FCMP)* function.
- Iteratively refine l and u based on FCMP outputs, adjusting them dynamically to minimize information loss.
- Terminate when the optimal clipping bounds are found, ensuring minimal degradation in quantized model accuracy.

Boundary Refinement (BR)

Purpose: Optimizes clipping parameters for stable quantization using backpropagation.

- Refines learned bounds to minimize reconstruction loss.
- Uses straight-through estimation (STE) to fine-tune boundaries.

Loss Function:

$$L_{\text{rec}} = \frac{1}{B} \sum_{i=1}^B \|X_i - \hat{X}_i\|_1 \quad (3)$$

where X_i is the original activation, $\hat{X}_i$ is the quantized activation, and B is the batch size.

Frequency-Guided Optimization (FGO)

Purpose: Ensures high-frequency details remain intact post-quantization.

- Computes Fourier Transform (FFT) before and after quantization.
- Minimizes frequency-aware error to preserve details.
- This helps analyze how much high-frequency information is lost due to quantization.

Mathematical Formulation:

$$\gamma = \frac{1}{C} \sum_{i=1}^C |||F_X| - |F_{\hat{X}}||| \quad (4)$$

where F_X is the FFT of the original activation, and $F_{\hat{X}}$ is the FFT of the quantized activation.

Appendix: Scale and Zero Point Calculation

Converting Floating-Point to Integer:

$$\text{scale} = \frac{u - l}{2^b - 1} \quad (5)$$

$$\text{zero point} = \text{clip} \left(\left\lfloor \frac{-l}{\text{scale}} \right\rfloor, 0, 2^b - 1 \right) \quad (6)$$

- l, u are the minimum and maximum bounds of the input.
- Scale maps floating-point values into an integer range.
- Zero point ensures correct alignment.

Recovering Floating-Point Values:

$$\hat{X}_i = \text{scale} \cdot (X_i^{\text{int}} - \text{zero point}) \quad (7)$$

- Converts quantized integers X_i^{int} back to an approximate floating-point representation.

Ensures Values Stay Within Range:

$$\text{clip}(x; a, c) = \begin{cases} a, & \text{if } x < a \\ x, & \text{if } a \leq x \leq c \\ c, & \text{if } x > c \end{cases} \quad (8)$$

- Limits values between a and c to avoid out-of-range errors.

The End